

Energy Transfer

Quick reference sheet · Physics · keep handy for revision

1. Energy stores

A system (an object, or group of objects) stores energy in eight possible ways:

Kinetic	Gravitational potential	Elastic potential	Thermal
anything moving	anything raised up	stretched/squashed springs	anything with a temperature
Chemical	Nuclear	Electrostatic	Magnetic
fuels, food, batteries	inside atomic nuclei	charges at a distance	magnets at a distance

2. Conservation, transfer & dissipation

Conservation of energy: energy cannot be created or destroyed, only transferred, stored, or dissipated. The total energy before a change equals the total energy after.

Energy is transferred along four pathways:

Mechanically — a force acting through a distance

Electrically — charge moving through a circuit

By heating — due to a temperature difference

By radiation — light or sound waves spreading out

Dissipation: "wasted" energy spreads into the surroundings (usually as heat) and becomes hard to use further — it is not destroyed, just spread out.

3. Kinetic & gravitational potential energy

kinetic energy = $\frac{1}{2} \times \text{mass} \times \text{speed}^2$ $E_k = \frac{1}{2}mv^2$

E_k in joules, J · m in kg · v in m/s. Doubling speed *quadruples* E_k (speed is squared).

gravitational potential energy = $\text{mass} \times g \times \text{height}$ $E_p = mgh$

E_p in J · m in kg · $g = 9.8 \text{ N/kg}$ near Earth's surface · h in m

4. Elastic potential energy & specific heat capacity

elastic potential energy = $\frac{1}{2} \times \text{spring constant} \times \text{extension}^2$ $E_e = \frac{1}{2}ke^2$

k in N/m · e in m · only valid up to the limit of proportionality

change in thermal energy = $\text{mass} \times \text{specific heat capacity} \times \text{temperature change}$ $\Delta E = mc\Delta\theta$

ΔE in J · m in kg · c in J/kg°C (a property of the material, e.g. water = 4200) · $\Delta\theta$ in °C

5. Power & efficiency

power = energy transferred ÷ time $P = E / t$ (measured in watts, W — 1 W = 1 J/s)

efficiency = useful output energy transfer ÷ total input energy transfer (×100 for a %)

No real machine is 100% efficient — some energy is always dissipated. Reduce unwanted transfers with **lubrication** (less friction) or **insulation** (less unwanted heating).

6. Worked examples

Given	Find	Working	Answer
$m = 2 \text{ kg}$, $v = 3 \text{ m/s}$	E_k	$\frac{1}{2} \times 2 \times 3^2$	9 J
$m = 5 \text{ kg}$, $h = 2 \text{ m}$	E_p	$5 \times 9.8 \times 2$	98 J
$k = 40 \text{ N/m}$, $e = 0.2 \text{ m}$	E_e	$\frac{1}{2} \times 40 \times 0.2^2$	0.8 J
$m = 2 \text{ kg}$, $c = 4200$, $\Delta\theta = 10^\circ\text{C}$	ΔE	$2 \times 4200 \times 10$	$8.4 \times 10^4 \text{ J}$
$E = 600 \text{ J}$, $t = 5 \text{ s}$	P	$600 / 5$	120 W
input = 500 J, useful = 350 J	efficiency	$350 / 500$	70%